

Confidence interval for the difference of two means. Independent samples, variance known

University example

Background You have the dataset from the lesson.

Task 1 Calculate the 99% confidence interval

Task 2 Compare it to the 95% confidence interval from the lesson

Solution:

	Engineering	Management	Difference
Size	100	70	?
Sample mean	58	65	-7.00
Population std	10	5	1.16

Task 1:

Z	CI low	CI high
99%	-10.01	-3.99

99% CI, $z_{0.005}$ 2.58

Task 2: The 95% confidence interval that we obtained in the lesson was [-9.28, -4.72]
A higher confidence leads to a broader interval (see Margin of error lesson).